

Notes: Bresnahan, Brynjolfsson, and Hitt (QJE, 2002)

Information Technology, Workplace Organization, and the Demand for Skilled Labor: Firm-Level Evidence

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1 Big picture

- **Question.** Does information technology raise the relative demand for skilled labour directly, or mainly through complementary changes in workplace organization, human capital policy, and product/service innovation?
- **Central idea.** The paper argues that the relevant form of skill-biased technical change is not simply computer capital by itself. The more complete technical change is a cluster of complements: information technology, decentralized workplace organization, and product/service innovation.
- **Core mechanism.** Declining IT prices lead firms to adopt more IT. But effective IT use requires organizational redesign, worker autonomy, teams, skill upgrading, training, and new output strategies. These complementary changes raise demand for skilled labour.
- **Empirical setting.** The authors combine a new survey of organizational practices and human capital at large firms with Computer Intelligence InfoCorp data on IT capital and Compustat data on output, employment, labour costs, and capital.
- **Main evidence.** Firms with more IT tend to have more skilled workers, more educated workers, more decentralized work practices, and more human capital investment.
- **Demand-side evidence.** Human capital and decentralized work organization predict IT demand, even after controlling for size, industry, production process, and other inputs.
- **Productivity evidence.** IT, human capital, and workplace organization are associated with higher measured productivity, especially when adopted together. Mixed combinations are often less productive than coherent clusters.
- **Managerial-belief evidence.** Managers in firms using more IT, skill, and decentralization report that computers increase skill needs and autonomy rather than deskill work.
- **Main takeaway.** The relationship between computers and skilled labour is mediated by organizational change. The paper's central contribution is to move the analysis from "IT versus skill" to a system of firm-level complements.

2 Incremental contribution of the paper

2.1 What the paper adds relative to earlier skill-biased technical change work

Earlier work had already documented a positive relation between computer use and wages or skills at the worker, firm, and industry levels. The incremental contribution of this paper is to open the black box behind that correlation. The authors ask *why* computers are associated with skilled labour and whether the answer lies in complementary organizational innovations.

- **From residual SBTC to a concrete mechanism.** The paper translates the broad idea of skill-biased technical change into a specific firm-level mechanism: IT-enabled organizational redesign.
- **From one input to a system of complements.** Instead of treating IT as an isolated technology, the paper studies the joint adoption of IT, decentralized workplace organization, human capital, and product/service innovation.
- **From industry data to firm-level practices.** The paper uses firm-level survey evidence on work organization, not just industry-level computer investment or worker-level computer use.
- **From correlation to multiple tests.** The authors do not rely on one correlation. They examine correlations, conditional input demand equations, production functions, human capital investment equations, and managerial beliefs.

- **From direct effects to interactions.** The paper emphasizes that the productivity and labour-demand effects of IT are stronger when IT is combined with skilled labour and decentralized work systems.

2.2 Why this contribution matters

- It changes the interpretation of the computer-skill correlation. Computers may not simply require more skill mechanically; they may make new organizational forms profitable, and those organizational forms require skill.
- It helps explain why the same IT may generate different outcomes across firms. Firms differ in their ability or willingness to reorganize.
- It implies that measuring only computer investment understates the relevant adjustment cost. Organizational redesign and human capital investment are part of the technology.
- It links the wage-inequality literature to the productivity and management-practices literature.

2.3 Incremental contribution in one sentence

The paper's incremental contribution is to show that IT-based skill-biased technical change is best understood as the adoption of a complementary system - IT, decentralized work practices, skilled labour, and output innovation - rather than as the independent effect of computers alone.

3 Data and measurement

3.1 Firm survey on organization and human capital

The authors conducted a survey of large U.S. firms to measure work organization and human capital practices at the firm and representative-establishment level.

- The survey universe is based on firms that have both Computer Intelligence InfoCorp IT data and Compustat accounting data.
- The final survey response rate is about 49 percent, yielding 379 responses.
- Roughly 55 percent of observations are in manufacturing, mining, or construction, and roughly 45 percent are in services.
- The survey asks about team-based organization, employee involvement, team-building, promotion for teamwork, breadth of jobs, and decision authority over pace and method of work.
- It also measures worker skill, education, training, screening, cross-training, and occupational composition.

Interpretation: The survey is crucial because decentralized workplace organization is not observable in standard datasets. The paper's ability to test complementarities comes from observing IT and organizational practices together.

3.2 Human capital measures

Human capital is measured in several ways:

- managers' assessments of worker skill and required education;
- shares of production workers with high school or less, some college, and completed college;
- occupational composition: unskilled blue-collar, skilled blue-collar, clerical, professional, and managerial;
- human capital investment policies: screening for education, training intensity, and cross-training.

Interpretation: The use of multiple human capital measures helps guard against the possibility that one noisy proxy drives the results. The paper finds similar patterns across education, skill assessment, and occupational structure.

3.3 Workplace organization measures

The main workplace organization variable, *WO*, summarizes decentralized and team-based work practices. It is constructed from variables including self-managing teams, quality circles, team-building, promotion for teamwork, job breadth, and worker decision authority over pace and method.

Interpretation: The organization measure captures the idea that IT-enabled firms move away from narrow, hierarchical task design toward broader jobs, more autonomy, and teams.

3.4 IT measures

The paper uses two sources of IT data.

- Computer Intelligence InfoCorp measures IT capital, total MIPS, and total PCs.
- The authors' organizational survey measures the degree of computerization of work, the percentage of workers using general-purpose computers, and the percentage using email.

Interpretation: The CII data provide independent firm-wide computer capital measures. The survey measures are closer to how computers are used by workers at the representative establishment.

3.5 Compustat production variables

The paper uses Compustat to obtain sales, value added, labour expense, non-IT capital, employment, and firm-level controls. These variables allow the authors to estimate production functions and compare productivity across firms.

Interpretation: Compustat links organizational and IT measures to economic outcomes. It also provides controls for scale and industry.

3.6 Product and service innovation

Product and service innovation is not measured directly. Instead, it is proxied through relative productivity or sales performance conditional on inputs.

Interpretation: If a firm uses IT and new work practices to create better products or services, this should appear as higher measured output or value added relative to observed labour and capital inputs.

4 Conceptual framework and empirical specifications

4.1 Complementarity system

The conceptual framework can be summarized as

$$\text{Decline in } P_{IT} \Rightarrow \text{more use of } \{IT, WO, S\} \Rightarrow \text{higher demand for skilled labour } \frac{L_h}{L_s}. \quad (1)$$

Here *IT* denotes information technology, *WO* denotes decentralized workplace organization, *S* denotes product/service innovation, and L_h/L_s denotes the ratio of high-skill to low-skill labour.

Interpretation: The paper's theory is not that IT alone causes all skill upgrading. IT makes a new organizational and product strategy profitable, and that strategy is skill intensive.

4.2 Short-run input choice equation

The conditional input choice equation is written as

$$Q_{it}^c = f(Q_{it}^{c'}, Controls), \quad (2)$$

where Q_{it}^c is firm i 's choice of input or technology c in year t , and $Q_{it}^{c'}$ denotes other hypothesized complements.

Interpretation: The more easily varied choice, such as IT capital or human capital investment, is modeled as a function of quasi-fixed complements such as skills or organization. A positive relation is interpreted as evidence of complementarity.

4.3 IT demand equation

The main IT demand regressions use specifications such as

$$\log(ITCAP_{it}) = \alpha + \beta_1 Skill_i + \beta_2 WO_i + \beta_3 \log(VA_{it}) + \beta_4 \log(Employment_{it}) + X_i' \gamma + \mu_j + \tau_t + \varepsilon_{it}. \quad (3)$$

Interpretation: If skilled labour and decentralized work are complements to IT, then β_1 and β_2 should be positive.

4.4 Human capital investment equation

The human capital investment regressions ask whether firms with more IT and decentralized organization invest more in human capital:

$$HKINVEST_i = \alpha + \beta_1 IT_i + \beta_2 WO_i + \beta_3 Skill_i + X_i' \gamma + \varepsilon_i. \quad (4)$$

Interpretation: A positive coefficient on IT or WO means firms with the complementary system also invest more in training and screening.

4.5 Production function

The production function is

$$\log(S_{it} - M_{it}) = f(L_{it}, K_{it}, Q_{it}; controls), \quad (5)$$

where S is sales, M is materials, so $S - M$ is value added. A standard empirical version is

$$\log(VA_{it}) = \alpha + \beta_L \log(Labor_{it}) + \beta_K \log(Capital_{it}) + \beta_{IT} IT_{it} + \beta_{HK} HK_i + \beta_{WO} WO_i \quad (6)$$

$$+ \beta_{IT \times HK} (IT_{it} \times HK_i) + \beta_{IT \times WO} (IT_{it} \times WO_i) + \mu_j + \tau_t + \varepsilon_{it}. \quad (7)$$

Interpretation: The level terms test whether IT, skill, and work organization are associated with higher output. The interaction terms test whether these inputs are more productive when used together.

4.6 Alternative explanations considered

- **Coincidence.** IT, skills, and organization may be correlated because of omitted industry or process differences.
- **Scale.** Large firms may demand more of every input.
- **Free cash flow or rents.** Successful firms may spend more on computers and skilled workers regardless of productivity.

- **Worker rents.** Skilled workers may receive computers as perks.
- **Fads.** Some managers may adopt IT, teams, and training together because they believe in a management fashion.
- **Omitted complements.** Software, management quality, or product innovation may also be part of the complementary system.

Interpretation: The paper’s strategy is to ask whether the same patterns appear in correlations, demand equations, productivity equations, and managerial belief measures. This triangulation is more convincing than any single regression.

5 Identification and empirical design

5.1 What identifies the empirical patterns

The paper exploits cross-firm variation in IT use, workplace organization, and human capital, along with firm-year variation in IT and Compustat variables.

- Cross-sectional variation identifies whether high-IT firms also have high skill and decentralized work practices.
- Panel variation identifies whether IT demand and production outcomes are related to organization and human capital.
- Industry, size, and production-process controls reduce confounding from broad technological or sectoral differences.

Interpretation: This is not a natural experiment. It is a carefully structured observational design aimed at distinguishing complementary production from simpler correlation stories.

5.2 Why multiple outcomes matter

- If IT and skills simply move together because of fashion, there need not be productivity effects.
- If IT and skill are complements, firms adopting them jointly should show higher productivity.
- If managers understand the complementarity, their stated beliefs should line up with their investment choices.

Interpretation: The paper’s credibility rests on the fact that the evidence points in the same direction across input demand, productivity, and managerial beliefs.

5.3 Limits of identification

- Organization and human capital are measured in a cross-section at the end of the sample period.
- Firms are not randomly assigned to IT or workplace practices.
- Some omitted factors, such as management quality or complementary software, could jointly drive IT, organization, skill, and productivity.
- The authors interpret some omitted complements as part of the broader technology system rather than as a threat to the central mechanism.

Interpretation: The paper establishes strong evidence of a complementary cluster but cannot fully separately identify all causal links inside that cluster.

6 Tables and figures

6.1 Tables I–III: data definitions and descriptive statistics

Read:

- Table I defines organization and human capital survey variables. It shows substantial variation in team practices, worker autonomy, education, occupation mix, and training.
- Table II defines the IT variables. The average firm has log IT capital of 3.07, mean computerization of work of 3.28 on a 1–5 scale, 53 percent of workers using general-purpose computers, and 31 percent using email.
- Table III defines production-function variables. The Compustat cross section has 311 firms with sales, value added, labour expense, non-IT capital, employment, and IT capital.

Economic message: The data section is the foundation of the paper’s contribution: it jointly observes IT, organization, human capital, and production outcomes for large firms.

TABLE 1
ORGANIZATIONAL PRACTICE AND HUMAN CAPITAL SURVEY VARIABLES

	Range	Variable	N	Mean	Std. dev.
<i>Variables measuring organization</i>					
Team-based work organization					
Use of self-managing teams	1–5	<i>SMTEAM</i>	345	2.11	1.13
Use of employee involvement groups	1–5	<i>QUALCIR</i>	345	2.85	1.21
Use of team-building activities	1–5	<i>TEAMBLD</i>	345	2.95	1.17
Promote for teamwork	1–5	<i>PROMTEAM</i>	345	3.59	0.95
Breadth of jobs	1–5	<i>BROAD</i>	345	3.25	0.99
Individual decision authority					
Who decides pace of work (3 = workers)	1–3	<i>PACE</i>	345	1.33	0.37
Who decides method of work (same)	1–3	<i>METHOD</i>	345	1.39	0.38
<i>Variables measuring human capital (levels)</i>					
Manager’s assessments 1–5					
Skill level of work	1–5	<i>SKILL</i>	345	3.60	0.86
Education level	1–5	<i>EDUC</i>	345	2.48	0.66
Education					
Workers w/ high school or less	0–100%	<i>%HSED</i>	263	59.3%	27.8%
Workers with some college	0–85%	<i>%SCED</i>	263	23.3%	17.5%
Workers completed college	0–100%	<i>%COLL</i>	263	17.4%	21.0%
Occupation mix					
Unskilled blue-collar (%)	0–95%	<i>%US</i>	337	18.4%	21.4%
Skilled blue-collar (%)	0–85%	<i>%SK</i>	337	24.7%	21.1%
Clerical (%)	0–80%	<i>%CL</i>	337	19.4%	17.6%
Professionals (%)	0–90%	<i>%PF</i>	337	20.7%	16.8%
Managers (%)	0–50%	<i>%MG</i>	337	16.8%	8.5%
<i>Variables measuring human capital (investment)</i>					
Pre-employment screen for education	1–5	<i>SCNED</i>	345	3.31	0.89
Training (% workers involved)	0–100%	<i>TRAIN</i>	345	48.0%	36.1%
Cross-train workers	1–5	<i>XTRAIN</i>	345	3.16	0.98

Source. Authors’ Survey.

Table I: Organizational practice and human capital survey variables

IT VARIABLES				
	Variable	N	Mean	Std. dev.
CII Survey*				
Log(IT Capital)	<i>LITCAP</i>	333	3.07	1.66
Total MIPS (Millions of instructions/sec.)	<i>MIPS</i>	333	2624	8737
Total PCs	<i>TOTPC</i>	333	4560	10,997
Organizational Survey**				
Degree of computerization of work 1–5	<i>COMP</i>	343	3.28	1.11
% Workers using general purp. computers	<i>%GP</i>	290	53.0%	33.8%
% Workers using email	<i>%EMAIL</i>	290	31.0%	32.2%

Table II: IT variables

TABLE III PRODUCTION FUNCTION VARIABLES				
	Variable	N	Mean	Std. dev.
<i>Computat variables</i>				
1994 Cross section*				
log(Sales)	<i>LSALES</i>	311	7.72	1.00
log(Value added)	<i>LVA</i>	311	6.79	1.02
log(Labor expense)	<i>LLABOR</i>	311	6.17	1.09
log(Non-IT capital)	<i>LNITCAP</i>	311	7.43	1.44
log(Employment)	<i>LEMPLOY</i>	311	2.55	1.09
log(IT Capital)	<i>LITCAP</i>	311	2.60	1.49
<i>Production function controls**</i>				
Production worker composition				
Blue-collar (fraction of jobs listed)	<i>PRBL</i>	345	61.9%	46.2%
Clerical (fraction of jobs listed)	<i>PRCL</i>	345	31.4%	43.4%
Professional (fraction of jobs listed)	<i>PRPF</i>	345	4.6%	17.5%

Table III: production function variables

6.2 Tables IV and V: correlations among IT, human capital, and workplace organization

Read:

- Table IV shows positive partial Spearman correlations among IT capital, workplace organization, worker skill, college education, and professional employment.
- The correlation between work organization and worker skill is 0.28 and statistically significant.
- The correlation between work organization and college education is 0.34 and statistically significant.
- Table V shows that multiple IT measures are positively related to skill, education, professionals, training, and screening.
- The survey-based IT measures have especially strong correlations with human capital variables.

Economic message: The complements move together. Firms with more IT are also firms with more skilled workers and more decentralized work systems.

TABLE IV CORRELATIONS BETWEEN MEASURES OF IT, HUMAN CAPITAL, AND ORGANIZATION					
Measure	Computer capital: <i>ITCAP</i>	Work organization: <i>WO</i>	Employee skill: <i>SKILL</i>	% College educated: <i>%COLL</i>	% Professionals: <i>%PF</i>
Computer capital (<i>ITCAP</i>)	1				
Work organization (<i>WO</i>)	.18***	1			
Worker skill (<i>SKILL</i>)	.12*	.28***	1		
Percent college (<i>%COLL</i>)	.05	.34***	.17***	1	
Percent professional (<i>%PF</i>)	.30***	.21***	.06	.21***	1

Spearman partial rank order correlations controlling for industry (nine sector dummy variables), employment (*EMPLOY*) and production worker composition (*PRBL, PRCL*). N = 251–401, due to nonresponse

Table IV: correlations among IT, human capital, and organization

TABLE V CORRELATIONS BETWEEN IT AND HUMAN CAPITAL						
Measure (scale in parentheses)	% Use gen. purpose computing: <i>%GP</i>	% Use email: <i>%EMAIL</i>	Computerization of work: <i>COMP</i>	Computer capital: <i>ITCAP</i>	Processing power: <i>MIPS</i>	Number of PCs: <i>TOTPC</i>
<i>Skills/education (N = 371)</i>						
Skill levels (<i>SKILL</i>)	.20***	.29***	.36***	.09	.15***	.13**
Education (<i>EDUC</i>)	.15**	.26***	.34***	.13	.08	.01
<i>Education Distribution (N = 237)</i>						
High school education (<i>%HSED</i>)	-.28***	-.37***	-.32***	-.11*	-.18***	-.15**
College graduate (<i>%COLL</i>)	.27***	.36***	.28***	.07	.14	.05
<i>Workforce composition (N = 303)</i>						
Clerical (<i>%CL</i>)	-.02	.04	-.05	-.08	-.01	-.03
Unskilled blue-collar (<i>%US</i>)	-.22***	-.25***	-.17***	-.08	-.13**	-.09
Skilled blue-collar (<i>%SK</i>)	-.05	-.05	.05	.05	.09	.02
Managers (<i>%MG</i>)	.16**	.19**	.11*	.17**	.15**	.10
Professionals (<i>%PF</i>)	.16**	.27***	.18**	.29***	.39***	.28***
<i>HR investment policies (N = 370)</i>						
Training (<i>TRAIN</i>)	.17***	.14**	.20***	.14**	.15***	.14***
Screen for education (<i>SCNED</i>)	.11*	.15***	.28***	.16***	.18***	.21***
Cross-training of workers (<i>XTRAIN</i>)	.18***	.07	.07	.02	.02	.03
<i>HR investment policies (N = 370) (control for SKILL and EDUC)</i>						
Training (<i>TRAIN</i>)	.14**	.10*	.19***	.15***	.14***	.14***
Screen for education (<i>SCNED</i>)	.07	.08	.15**	.16***	.15***	.19***
Cross-training of workers (<i>XTRAIN</i>)	.18***	.07	.03	.01	-.05	.01

Spearman partial rank order correlations controlling for industry (nine sector dummy variables), employment (*EMPLOY*) and production worker composition (*PRBL, PRCL*). N = 240–370, due to nonresponse and some measures limited to second and third wave surveys.

Rep: * $p < .1$, ** $p < .05$, *** $p < .01$, test is against the null hypothesis that the correlation is zero.

Table V: correlations between IT and human capital

6.3 Table VI: IT demand and complementary inputs

Read:

- The dependent variable is log IT capital.
- Worker skill positively predicts IT demand in column (1): a one-standard-deviation increase in skill is associated with about 14.2 percent more IT capital.

- Work organization is also positive and significant in column (2).
- When education and work organization are both included, work organization remains positive while the coefficient on worker skill or education changes.
- The table controls for value added, employment, sector dummies, workforce composition, and year effects.

Economic message: IT demand is higher when firms have skilled workers and decentralized work practices. This is direct evidence that IT is complementary with both human capital and organization.

TABLE VI
IT DEMAND OVER 1987–1994 AS A FUNCTION OF HUMAN CAPITAL AND
WORKPLACE ORGANIZATION AT THE END OF THE SAMPLE PERIOD

Specification	Computer capital: log(ITCAP)	Computer capital: log(ITCAP)	Computer capital: log(ITCAP)	Ordinary capital: log(NITCAP)
Variable	WO	Education + WO	Education + WO	
	(1)	(2)	(3)	(4)
Worker skill (<i>SKILL</i>)	.142** (.0559)	.0915 (.0608)	.0756 (.0611)	.0464 (.0681)
College education (% <i>COLL</i>)			.0108* (.0600)	-.0745 (.0711)
Some college (% <i>SCED</i>)			-.0174 (.0573)	.0122 (.0603)
Decentralization (<i>WO</i>)		.142** (.0642)	.123* (.0641)	.0116 (.0641)
log(Value-added)	.714*** (.115)	.709*** (.115)	.668*** (.123)	.820*** (.223)
log(VA)				
log(Employment)	.104 (.128)	.0857 (.130)	.128 (.131)	.0486 (.214)
log(<i>Employ</i>)				
Controls: sector dummies, year dummies, workforce composition (<i>PRCL</i> , <i>PRPF</i>)	Controls	Controls	Controls	Controls
R^2	49.3%	50.2%	50.6%	62.3%
N	1331	1331	1331	1331

Key: *— $p < .1$, **— $p < .05$, ***— $p < .01$.
Huber-White robust (clustered by firm) standard errors utilized to account for repeated observations for same firm. Largest common sample used—results similar in maximal sample for each regression (not shown).

TABLE VII
RELATIONSHIP BETWEEN HUMAN CAPITAL INVESTMENT AND VARIOUS MEASURES OF
INFORMATION TECHNOLOGY, WITH CONTROLS FOR SKILL AND WORKPLACE
ORGANIZATION

Dependent variable	Human capital investment	Human capital investment	Human capital investment	Human capital investment	Human capital investment	Human capital investment
Specification	OLS	OLS	IV	IV	OLS	OLS
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Computerization log(ITCAP/EMPLOY) ₋₄	.180*** (.0673)	.154*** (.0614)				
Computerization log(ITCAP/EMPLOY)			.184** (.0735)			
Computerization (<i>COMP</i>)				.994* (.522)		
Computerization log(MIPS/EMPLOY) ₋₄					.172*** (.0644)	
Computerization log(TOTPC/EMPLOY) ₋₄						.0812 (.0650)
Work organization (<i>WO</i>)		.419*** (.0589)	.409*** (.0594)	.314*** (.0982)	.403*** (.0607)	.449*** (.0569)
Skills (<i>SKILL</i>)	.237*** (.0629)	.0948 (.0607)	.0930 (.0609)	-.240 (.200)	.0911 (.0618)	.100 (.0626)
Industry controls	Sector dummies	Sector dummies	Sector dummies	Sector dummies	Sector dummies	Sector dummies
N	250	250	250	250	250	250

Key: *— $p < .1$, **— $p < .05$, ***— $p < .01$.
All variables are standardized to mean 0, unit variance.
IV: Computerization (*ITCAP/EMPLOY* and *COMP*) instrumented with fourth lagged log(ITCAP/EMPLOY); all other variables are considered exogenous.

6.4 Table VII: human capital investment and IT

Read:

- The dependent variable is human capital investment.
- Computerization measures positively predict training and screening investments.
- Work organization is positive and significant across multiple specifications.
- Instrumenting computerization with lagged IT gives similar results.
- Existing skill levels do not eliminate the relation between IT, work organization, and human capital investment.

Economic message: Firms with the IT-organization system invest more in workers. Skill upgrading is not just a pre-existing condition; it is part of the adjustment process.

6.5 Table VIII: productivity and interaction effects

Read:

- The dependent variable is log value added.

- IT stock is positive and significant across specifications.
- Interactions between IT and human capital are positive: the output elasticity of IT is higher in firms with skilled workers.
- Interaction estimates are economically meaningful but not always precisely estimated.
- The paper reports predicted productivity calculations showing high productivity for coherent high-IT, high-skill, high-organization firms.

Economic message: The productivity evidence argues against pure fad or rent stories. Firms that combine IT, skill, and decentralized organization perform better than firms with incoherent mixes.

TABLE VIII
PRODUCTIVITY EFFECTS OF COMPUTER-HUMAN CAPITAL AND COMPUTER-WORK ORGANIZATION-HUMAN CAPITAL INTERACTIONS FROM 1987-1994 USING HUMAN CAPITAL AND WORKPLACE ORGANIZATION METRICS AT THE END OF THE SAMPLE

Dependent variable	log(VA)	log(VA)	Log(VA)	log(VA)	log(VA)
Variable	Specification	Baseline	Baseline + skills	Baseline + skills (maximum sample)	Baseline + WO (maximum sample)
		(1)	(2)	(3)	(4)
IT stock		.0347** (.0175)	.0330** (.0168)	.0371*** (.0148)	.0358** (.0147)
log(ITCAP)		.136*** (.0298)	.140*** (.0295)	.157*** (.0247)	.144*** (.0246)
capital stock		.753*** (.0422)	.733*** (.0418)	.752*** (.0375)	.748*** (.0374)
log(LABOR)			.00187 (.0210)	.00244 (.0155)	
worker skill					.743*** (.0407)
SKILL					
college education					-.00795 (.0244)
%COLL					
T × skill			.0262 (.0195)	.0224* (.0155)	
log(ITCAP) × SKILL					.0550** (.0272)
T × college					
log(ITCAP) × %COLL				.0218* (.0129)	
Work Organization (WO)				.0162* (.00844)	
Work organization × IT					
WO × log(ITCAP)					
controls		Sector dummies, Year dummies 1331 90.9%	Sector dummies, Year dummies 1331 90.9%	Sector dummies, Year dummies 2255 92.9%	Sector dummies, Year dummies 1331 91.2%

Key: ** $p < .1$, *** $p < .05$, **** $p < .01$. Huber-White robust (clustered by firm) standard errors utilized to account for repeated observations for the same firm. Largest common sample used except for columns (3) and (4) that utilizes the full sample—results are similar in maximal sample for the other regressions (not shown).

TABLE IX
CORRELATIONS BETWEEN MANAGER'S PERCEPTION OF COMPUTER EFFECTS WITH COMPUTER USE, HUMAN CAPITAL AND DECENTRALIZATION (1-5 SCALE WITH 3 = "NO CHANGE")

Measure	Computers increase skill	Computers increase autonomy	Computers increase monitoring	Computers routinize work
Mean	4.17***	3.58***	3.59***	2.96
Computer capital (ITCAP)	.09	-.02	-.06	-.08
% Use computer (%GP)	.25***	.07	-.04	.00
% Use email (%E mail)	.20***	.12**	-.04	-.10*
Work organization (WO)	.19***	.13***	.03	.05
Worker skill (SKILL)	.27***	.20***	.12**	.04
Percent college (%COLL)	.19***	.03	-.07	-.12*
Human capital investment (HKINVEST)	.22***	.10*	-.01	-.01

Spearman partial rank order correlations controlling for industry (nine sector dummy variables), employment (EMPLOY) and production worker composition (PRBL, PRCL). N = 238 - 295.

6.6 Table IX: managers' beliefs about computers

Read:

- Managers on average report that computers increase skill needs, increase worker autonomy, and increase monitoring.
- The strongest correlation is between the belief that computers increase skill and actual IT, computer use, work organization, skill, college share, and human capital investment.
- Firms whose managers perceive complementarity are also the firms that adopt the complementary cluster.

Economic message: Managers' beliefs line up with the econometric evidence. Firms investing in IT, skills, and organization are not merely following random fads; managers report a perceived complementarity between computers and skilled work.

7 Short summary

- The paper studies whether IT is part of a broader skill-biased technical change.
- The key claim is that IT works together with decentralized organization, product/service innovation, and human capital.
- The authors build new firm-level data combining IT, organizational practices, human capital, and production variables.
- IT, workplace organization, and human capital are positively correlated across firms.
- Skilled labour and decentralized work organization predict IT demand.
- IT and workplace organization predict human capital investment.

- IT, skill, and organization are associated with higher productivity when adopted together.
- Managers in high-IT, high-skill firms believe computers increase skill requirements and autonomy.
- The evidence supports a complementarity view of IT-enabled organizational change.
- The paper's most important contribution is shifting the SBTC discussion from computers alone to coherent systems of workplace practices and technologies.

8 Conclusion

8.1 Core conclusion

Bresnahan, Brynjolfsson, and Hitt show that the demand for skilled labour associated with information technology is best understood through a system of complementary firm-level innovations. IT adoption is connected to decentralized workplace organization, human capital investment, and product/service innovation.

8.2 Economic intuition

Computers make information cheaper and more widely available inside firms. That changes the optimal organization of work: employees can exercise more autonomy, work in teams, and solve problems closer to production or customers. But this organization requires higher skill and training. Therefore IT raises skill demand partly by making decentralized, skill-intensive work systems more valuable.

8.3 Incremental contribution takeaway

The paper's incremental contribution is to show that skill-biased technical change is not merely embodied in machines. It is embedded in a coherent organizational system. This insight helps explain why firms differ in the returns to IT and why IT adoption is closely tied to workplace redesign and rising demand for human capital.